Conduction, convection, radiation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify mechanisms of heat transfer and support each classification.

SUCCESS CRITERIA - I can...

1. Describe contact transfer, bulk fluid motion and electromagnetic transfer.
2. Use specific details or measurements from the situation.
3. Recognize that more than one mechanism can act.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “conduction” and “convection” to explain your answer. Sentence starters are optional.

conduction: energy transfer through microscopic interactions in matter

convection: energy transfer through bulk motion of a fluid

radiation: energy transfer by electromagnetic waves

01 Warm-up: bring back an earlier idea

A simulated source has GM=**150 m³/s²**. At r=**3 m**, find escape speed and **name two things your calculation** treats as true.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $E_i + E_{in} = E_f + E_{out}$.
Account for energy across the system boundary. Conduction: matter interactions; convection: fluid motion; radiation: electromagnetic waves.

Worked reasoning: A metal rod transfers energy from its hot end toward its cool end by conduction. A circulating fluid can carry energy by convection; radiation can cross a vacuum.

Your turn: 1. **Draw a separate**, labeled **diagram** for each way energy is transferred.
2. **Show the direction of each** energy transfer.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Conduction	Convection	Radiation
Sketch:	Sketch:	Sketch:
_____	_____	_____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Classify: a metal spoon warming in soup; circulating water in a pot; sunlight warming your hand. **Give a reason** for each.

Sentence starter: I notice ... . This supports ... because ... .

Situation	Mechanism	Evidence
Spoon in soup	____	____
Circulating water	____	____
Sunlight on skin	____	____

Marzano check - circle: ① ② ③

04 Apply the physics

A wall transfers energy from a warm room to outdoors at a steady **40 W** for **5 minutes**.
1. **Label the direction** and **calculate the energy transferred**.
2. **Predict how doubling the wall** thickness affects conduction if material, area and temperature difference stay fixed.

Sentence starter: Both ... . They differ in ... because ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

An asteroid travels through nearly empty space. Which of these mechanisms can transfer thermal energy to it from sunlight? Why not ordinary air convection?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Supplied data: m=**0.4 kg**, drop h=**0.6 m**, final speed **2 m/s** from rest, g=**9.8**. **Find mgh**, translational K and the fraction appearing as translation.
Show the energy changes and transfers; **label units**.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A hot object in a vacuum still cools. **Name the transfer mechanism** and **explain whether** absolute zero surroundings are required.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A vacuum flask still loses heat. **Name two paths** that remain and propose measurements that distinguish them.

Sentence starter: I notice … . This supports … because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3